

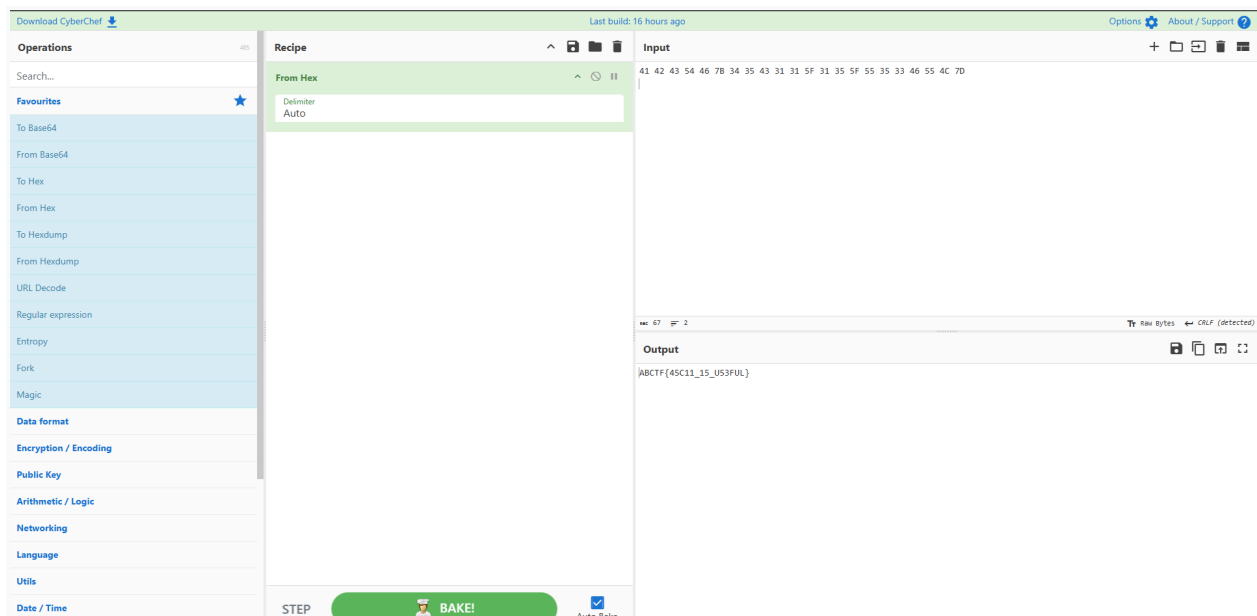
Title: Cryptography

Problem-In the computing industry, standards are established to facilitate information interchanges among American coders. Unfortunately, I've made communication a little bit more difficult. Can you figure this one out? 41 42 43 54 46 7B 34 35 43 31 31 5F 31 35 5F 55 35 33 46 55 4C 7D

Initial Analysis: Firstly i can see the numbers but there i can also see some characters like A,C,D,F from there i assumed that it could be a HEX. It means the plaintext is HEXED.

Tools Used: <https://gchq.github.io/CyberChef/>

Solution: Firstly i went to my browser and opened CyberChef , there I pasted the HEXED text and and Used the From Hex option and then i decoded the HEXED text.



Flag: ABCTF{45C11_15_U53FUL}

Takeaways: Pattern Recognition: Hexadecimal strings are easily identifiable by their 0-9 and A-F character set. When you see pairs of characters in this range, ASCII encoding should be your first guess. **Tooling is Everything:** CyberChef (the "Swiss Army Knife") is the industry standard for rapid prototyping of decodings. The "Magic" ingredient can also help if you're unsure of the format.

Problem-At 1pm I called my uncle who was 64 years old 10 months ago, but I heard only that. Later I started thinking about the 24 hour clock.

I hope you will help me solve this problem. Got a wav audio file.

Initial Analysis:Firstly when I opened the audio file I heard the number typing sound. It was following a pattern and it hid the message in the number typing thing. First i needed to decode the number pattern

Tools Used:DTMF Decoder, CyberChef

Solution:First i put the audio file in the DTMF Decoder and it decoded the string and i got this string Decoded: 67847010810197110123678289808479718265807289125. Then it seemed like an ASCII value to me so I went straight to the cyberchef and put that value in there but cyberchef was struggling but I had to add spaces to make the Ascii values more accurate after that i got my desired flag from there.

DTMF Decoder

Audio file:

Choose file

you_know_what_to_do.wav

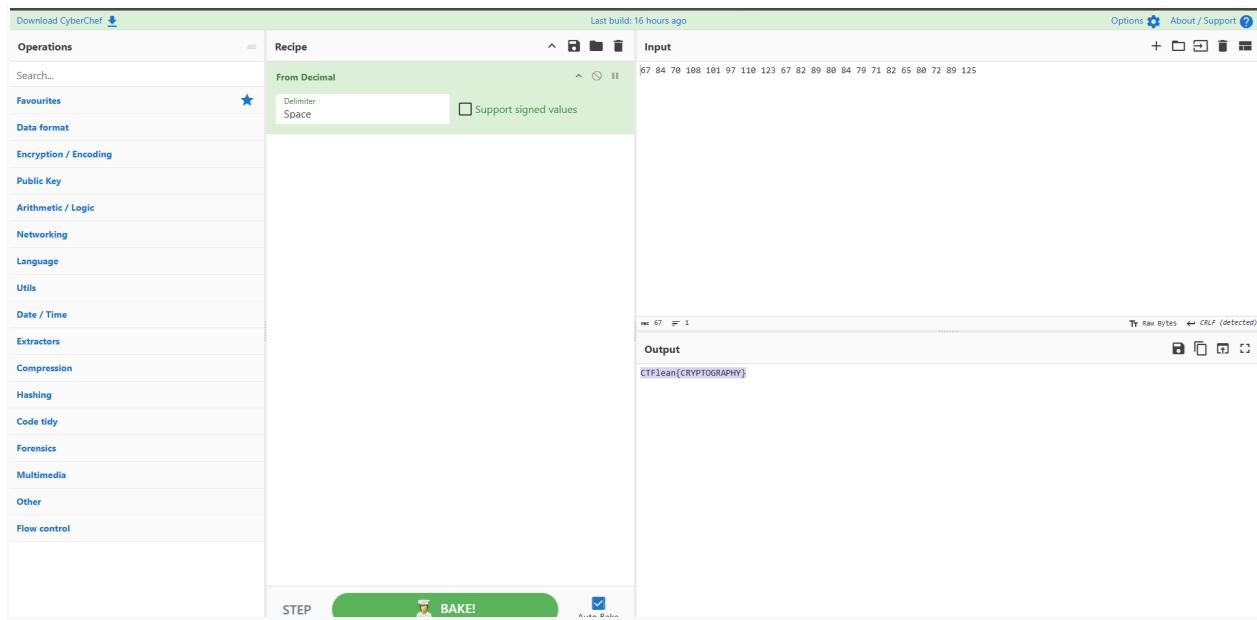
Sensitivity threshold:

0.050

Decode

Output

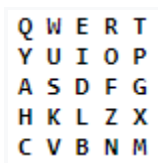
```
0001s 666..777.....
0002s .....888..444.....
0003s .....777..
0004s 000.....
0005s .....111..000..888.....
0006s .....111..
0007s 000..111.....
0008s .....999..777.....
0009s .....111..
0010s 111..000.....
0011s .....111..222..333..
0012s .....
0013s 666..777.....
0014s .....888..222.....
0015s .....888..
0016s 999.....
0017s .....888..000.....
0018s .....888..444..
0019s .....
0020s 777..999.....
0021s .....777..111.....
0022s .....888..
0023s 222.....
0024s .....666..555.....
0025s .....888..000..
0026s .....
0027s 777..222.....
0028s .....888..999.....
0029s .....111..
0030s 222..555..
Decoded: 67847010810197110123678289808479718265807289125
```



Flag: CTFlearn{h4ck3d}

Takeaways: Decimal Squashing: When a DTMF decoder gives you a giant wall of numbers with no spaces, it's usually Decimal ASCII. Look for "anchor" numbers like 123 (}) and 125 (}) to confirm. Manual Splitting: If tools fail because of the lack of spaces, look for the \$1xx\$ values (like 108, 110, 123) first, then the 2-digit values around them. CTF Archetype: If you hear "telephone beeps," think: DTMF → Decimal → ASCII.

Problem-Hello! My friend Fari sent me this suspicious message: 'MQDzqdor{Ix4Oa41W_1F_B00h_m1YlqPpPP}' and photo.png. Help me decrypt this!



Initial Analysis: I couldn't understand what type of cipher it was but after reading some blogs i got to know that it is a playfair-cipher.

Tools Used: <https://www.boxentriq.com/ciphers/playfair-cipher>

Solution: I went to the website and entered the cipher text MQDzqdor{Ix4Oa41W_1F_B00h_m1YlqPpPP} and for the key i placed the images writing QWERTYUIOPASDFGHKLZXCVBNM and it was decoded in no time.

BOXENTRIQ Workspace Naviga

On this page

- Start
- Playfair Cipher
- Features
- What is the Playfair cipher?
- History and facts
- Where You Will Encounter It

Playfair Cipher

Ⓜ Your data stays local. [Feedback](#)

Language: English

Non-English letters will be translated to English (for example Å → AA).

Ciphertext or plaintext Options

MQDzqdor{Ix4Oa41W_1F_B00h_m1YlqPpPP}

Mode: Decrypt Key: QWERTYUIOPASDFGHKLZXCVBNM

Encrypt = plaintext to ciphertext. Decrypt = ciphertext to plaintext. Duplicates are removed and only letters from the selected alphabet are used.

Merge letters for the 5x5 square: J to I Padding letter: X Duplicate letters: Split duplicates

Playfair uses 25 letters. English typically merges J into I. Used to split duplicates and pad the last pair. Applies when encrypting only.

Result Options

CTFlearn{P14Yf41R_1S_C00l_c1PheRrRR}

▼ Key square

The square is rebuilt as you change the key and merged letter.

Keyed alphabet: QWERTYUIOPASDFGHKLZXCVBNM

Q	W	E	R	T
Y	U	I	O	P
A	S	D	F	G
H	K	L	Z	X
C	V	B	N	M

Flag:CTFlearn{P14Yf41R_1S_C00l_c1PheRrRR}

Takeaways: The 5*5 Grid: Playfair uses a 5*5 matrix based on a keyword. Because $5*5 = 25$, one letter is always omitted or combined (usually I and J are treated as the same character).

Digraph substitution: Unlike Caesar or Vigenère, Playfair encrypts pairs of letters (digraphs). This means the ciphertext length will almost always be even.

No Double Letters: A major "tell" for Playfair is that a pair of letters in the ciphertext will never consist of the same letter (e.g., you'll never see LL or EE as a pair).

The "X" Padding: If you see a lot of 'X's or 'Q's scattered in the plaintext, they were likely used as "fillers" to separate double letters (like HELXLO) or to complete an odd-numbered string.

Geometric Rules:

- Same Row: Shift left to decrypt.
- Same Column: Shift up to decrypt.
- Rectangle: Swap corners.

Problem-One of the easiest and earliest known ciphers but with XXI century twist! Nobody uses Alphabet nowadays, right? Why should you when you have your keyboard?

BUH'tdy,|Bim5y~Bdt76yQ

Initial Analysis: Why should you when you have your keyboard? This was a hint I got when I first read the story and after some research I got to know that it is indeed a keyboard shift cipher.

Tools Used:<https://www.dcode.fr>

Solution:First i went to the [dcode.fr](https://www.dcode.fr) website and searched for keyboard shift cipher and pasted the cipher and hit decode that gave me the flag for this problem.

Search for a tool

★ SEARCH A TOOL ON dCODE
e.g. type 'sudoku'

★ BROWSE THE FULL dCODE TOOLS LIST

Results

BUH'tdy,|Bim5y~Bdt76yQ

↑↓	↑↓
qwerty ->2	CTFlearn{Cyb3rCae54r}
qwerty ->,-2	COFaegr/{My.3iCge94i}
qwerty ->2	MOKaugi/wMp.7iMgu98iE
dvorak ->3	JAU/ie,=LJok2,Jei43,V
colemak U	DYKophimJK.kb;5tp8ju1
qwerty U2	TT^aban/+C,.grTgb5hiZ
colemak U	DLKapti. V.,bu5hp6j;1
qwerty ->,-2	MTKluaiñwCpb7rMau58rE
dvorak ->5	:PA'ed1,C0sm'y:de7[yM
qwerty U2	TO^1bgnn+M,bgiTab9hrZ
dvorak ->,-3	JHUpin,yLVoz2cJni03cV
dvorak ->,-2	W0N=sug\<Khx7.Wus58.K
dvorak ->J4	ZUE'ad',PBam9yQda72yB
dvorak ->U4	ZUE'ad',PBam9yQda72yB
dvorak ->,-3	VAS/-ec=>Jtk8,Ve-49,X
dvorak ->5	:NAfo-1gY:sq01:-o] [TM
qwerty ->J4	>U5-odw,RBrm9yZdo72yT
dvorak ->J3	JPS'idc,L0tm2yVdi79yV
qwerty U2	TO^1bgnn}M,bgiTab9hrZ

dCode is preparing a new interface. Come test and give your feedback on the new page: [Keyboard Shift Cipher!](#)

KEYBOARD SHIFT CIPHER

Cryptography > Substitution Cipher > Keyboard Shift Cipher

GoDaddy Build, brand, and back your business with GoDaddy

KEYBOARD SHIFT DECODER

★ KEYBOARD SHIFTED CIPHERTEXT
BUH'tdy,|Bim5y~Bdt76yQ

★ PLAINTEXT EXPECTED LANGUAGE English

★ KEYBOARD LAYOUT Automatic Detection

★ SELECTED SHIFT Automatic Detection

★ NUMBER OF KEYS/STEPS FOR SHIFT Automatic Detection

★ USE ONLY ALPHANUMERIC CHARACTERS (NO SYMBOLS) ☐

▶ DECRYPT

See also: [Shift Cipher](#) – [Keyboard Coordinates](#) – [Caesar Cipher](#)

ALTERNATIVE METHOD: [KEYBOARD CHANGE](#)

▶ Go to: [Keyboard Change Cipher](#)

KEYBOARD SHIFT ENCODER

★ KEYBOARD SHIFT PLAIN TEXT
dcode keyboard

★ KEYBOARD LAYOUT QWERTY (US)

Summary

- ★ [Keyboard Shift Decoder](#)
- ★ [Keyboard Shift Encoder](#)
- ★ [What is a keyboard shift cipher? \(Definition\)](#)
- ★ [How to encrypt using Keyboard Shifting cipher?](#)
- ★ [How to decrypt Keyboard Shifting cipher?](#)
- ★ [How to recognize Keyboard Shifting ciphertext?](#)
- ★ [What are the variants of the Keyboard Shifting cipher?](#)

Similar pages

- ★ [Caesar Cipher](#)
- ★ [Keyboard Coordinates](#)
- ★ [Shift Cipher](#)
- ★ [Keyboard Change Cipher](#)
- ★ [Numeric Keypad Draw](#)
- ★ [ALT-Codes](#)
- ★ [Alphabetical Ranks Added](#)
- ★ [DCODE'S TOOLS LIST](#)

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Forum/Help

[DISCORD](#)

Flag:CTFlearn{Cyb3r_Cae54r}

Takeaways:The "QWERTY" Clue: If the ciphertext looks like gibberish but follows a physical path on your keyboard (e.g., rdgh instead of tfgh), suspect a keyboard shift.

Directional Offsets: Letters are typically shifted left, right, up, or down from their original position. For example, a "shift right" would turn H into J or E into R.

The "I" and "O" Edge Case: Remember that shifts wrap around or hit the "edge" of the keyboard (like P shifting right into nothing, or wrapping back to Q).

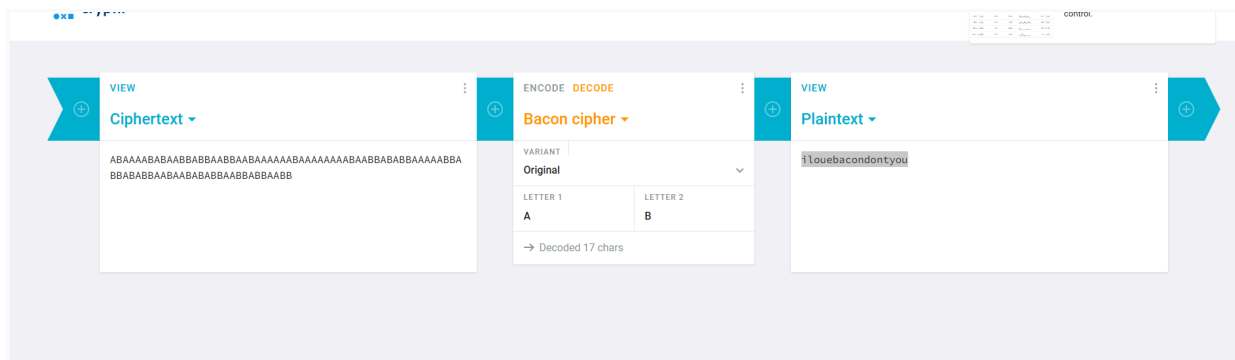
Problem-I love the smell of bacon in the morning!

ABAAAABABAABBABBAABBAABAAAAABAAAAAABAABBABABBBAAAAABB
ABBABABBAABAABABABBAABBABBAABB

Initial Analysis:First its a mix of A's and B's and we can find the hint in the line I love the smell of bacon in the morning! So I will think of the bacon cipher.

Tools Used:<https://cryptii.com>

Solution: I went to the crypti website and selected the Bacon cipher and changed the mode to DECODE and in the cipher text and pasted the text and changed the letters to A and B and it decoded the cipher in no time.



Flag:CTFlearn{ilouebacondontyou}

Takeaways:Binary in Disguise: At its core, it's a substitution cipher where every letter is replaced by a string of five characters using only two options (usually A and B).

The "Rule of 5": If your ciphertext length is a multiple of 5 (e.g., 25, 50, 100 characters), Bacon is a prime suspect.

Problem: Meet ROXy, a coder obsessed with being exclusively the worlds best hacker. She specializes in short cryptic hard to decipher secret codes. The below hex values for example, she did something with them to generate a secret code, can you figure out what? Your answer should start with 0x.

0xc4115 0x4cf8

Initial Analysis: We can see how ROXy has been named. This is a clue to use XOR operation to generate a secret Code.

Tools Used: <https://xor.pw/#>

Solution: I went to the XOR calculator I mentioned above then put those input values in the input and I got the calculated XOR value. After that I built the flag.

XOR Calculator

Thanks for using the calculator. [View help page.](#)

I. Input: [hexadecimal \(base 16\)](#) ▼

c4115

II. Input: [hexadecimal \(base 16\)](#) ▼

4cf8

Calculate XOR

III. Output: [hexadecimal \(base 16\)](#) ▼

c0ded

Warning:

Hexadecimal variable was not valid base 16. Updated input value may differ from what you provided.

[Home](#)

[Help](#)

[Privacy](#)

Flag:CTFlearn{0xc0ded}

Takeaway: The Reversibility Trick: XOR is its own inverse. If $A \text{ XOR } B = C$, then $C \text{ XOR } B = A$. You use the exact same key/process to both encrypt and decrypt.

Problem: I got a new hard drive just to hold my flag, but I'm afraid that it rotted. What do I do?

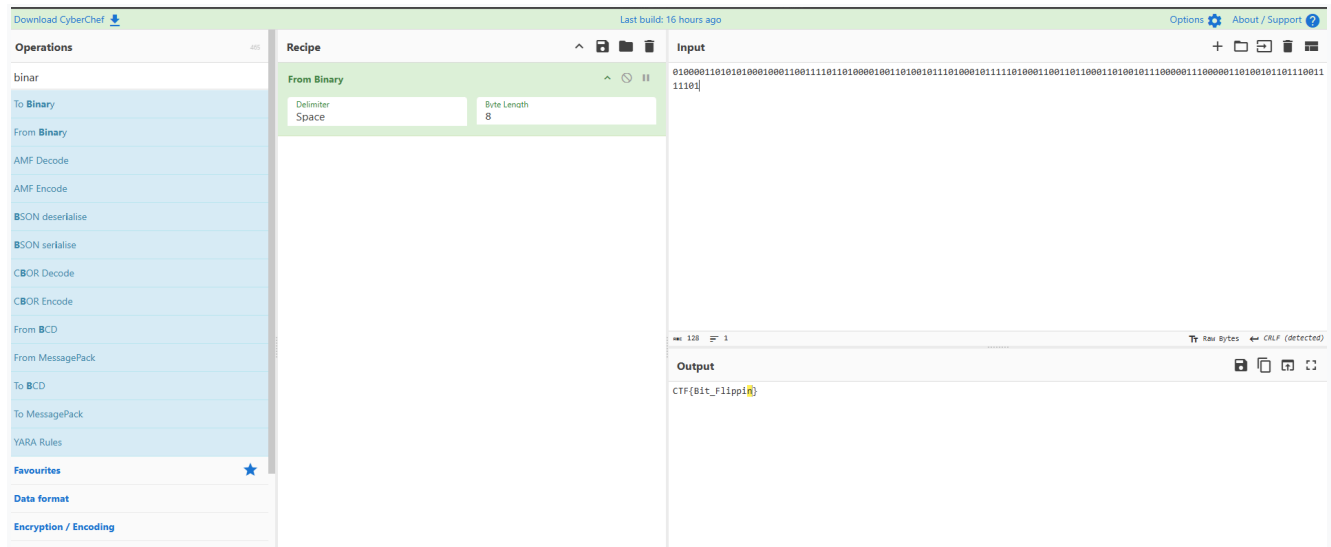
The only thing I could get off of it was this:

```
0100001101010100010001100111101101000010011010010111010001011111010001100110110  
0011010010111000001110000011010010110111001111101
```

Initial Analysis: It's clearly visible that it is a binary encoding.

Tools Used: CyberChef

Solution: So I went to the cyberchef online tool and I pasted the binary string in the input and selected the From binary and it decoded the string.



Flag: CTF{Bit_Flippi}

Takeaway: The Rule of 8: Standard ASCII characters are 8 bits (1 byte). If your string length isn't a multiple of 8, check for missing leading zeros or see if it's 7-bit ASCII.

The "01" Tell: In the ASCII table, most printable English characters start with 01 (uppercase starts with 010, lowercase with 011). If you see that pattern repeating every 8 bits, you're on the right track.

Non-Standard Binary: Binary isn't always 0s and 1s. It can be hidden as any two opposites:

Short/Long sounds, High/Low voltages, or Even/Odd numbers.

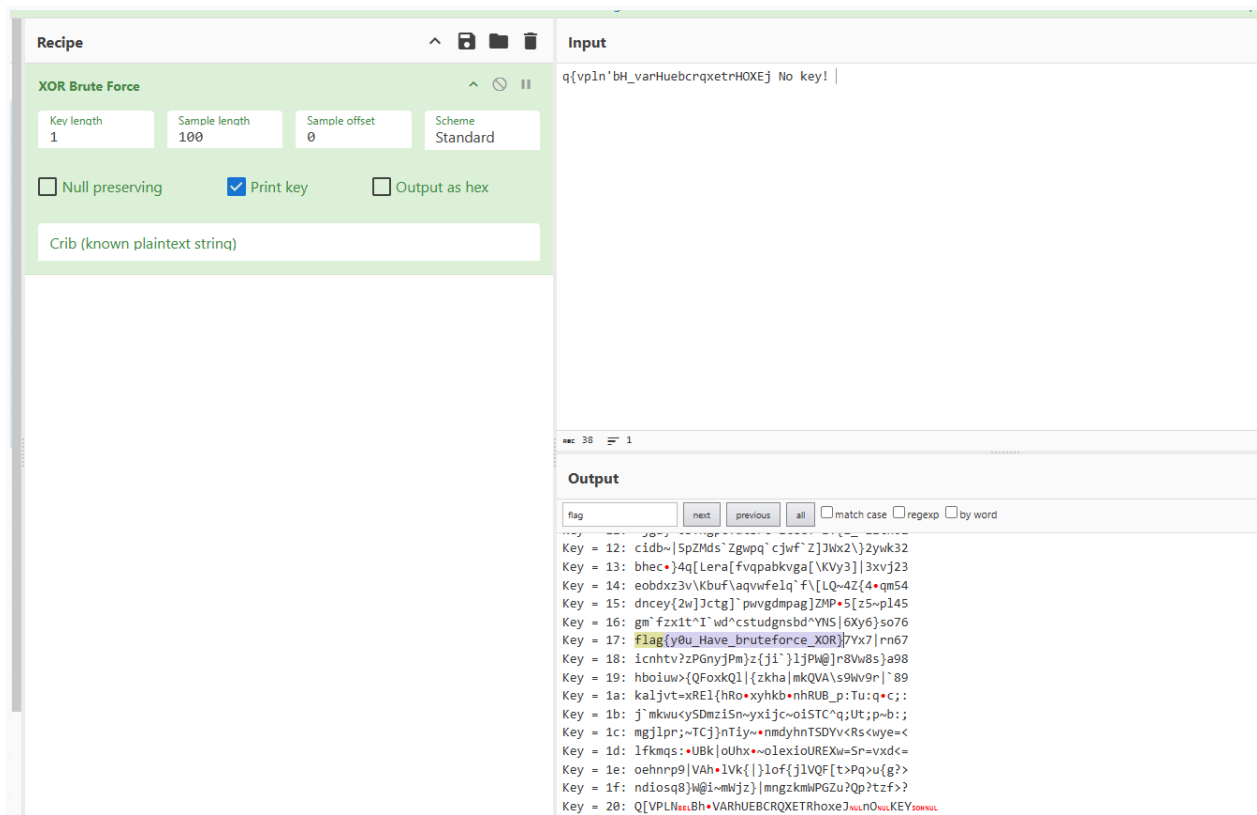
Bit Inversion: If "From Binary" gives you garbage, try a NOT operation (flipping all 0s to 1s) before decoding. Sometimes the logic is simply reversed.

Problem: There is a technique called bruteforce. Message: q{vp1n'bH_varHuebcrqxetrHOXEj
No key! Just brute .. brute .. brute ... :D

Initial Analysis: The problem name gives a hint BruXOR so i will do this.

Tools Used: CyberChef

Solution: I went to the cyberchef and selected XOR BRUTE FORCE and pasted the string in the input. It gave me a lot of output and I grabbed the flag from the output box.



Flag:CTFlearn{y0u_Have_bruteforce_XOR}

Takeaway: The 256-Key Limit: A single-byte XOR key only has 256 possibilities (0x00 to 0xFF). This makes it the "low-hanging fruit" of CTF crypto; always try this first if you suspect XOR.

Search for "Anchors": Don't read through all 256 results manually. Use a "Find" or "Grep" for known flag formats like flag{, CTF, or even just the { character.

The CyberChef "Sample" Trick: When using the XOR Brute Force recipe, you can set a "Search String." This filters the output and only shows you the keys that produce your specific keyword (like "flag").

Beyond the Single Byte: If a single-byte brute force fails, the key might be 2 or 3 bytes. The "keyspace" grows exponentially, but for small keys (up to 4 bytes), tools like xortool can still crack it by looking for the most frequent character (usually a space or null byte).

Problem: The vigenere cipher is a method of encrypting alphabetic text by using a series of interwoven Caesar ciphers based on the letters of a keyword.

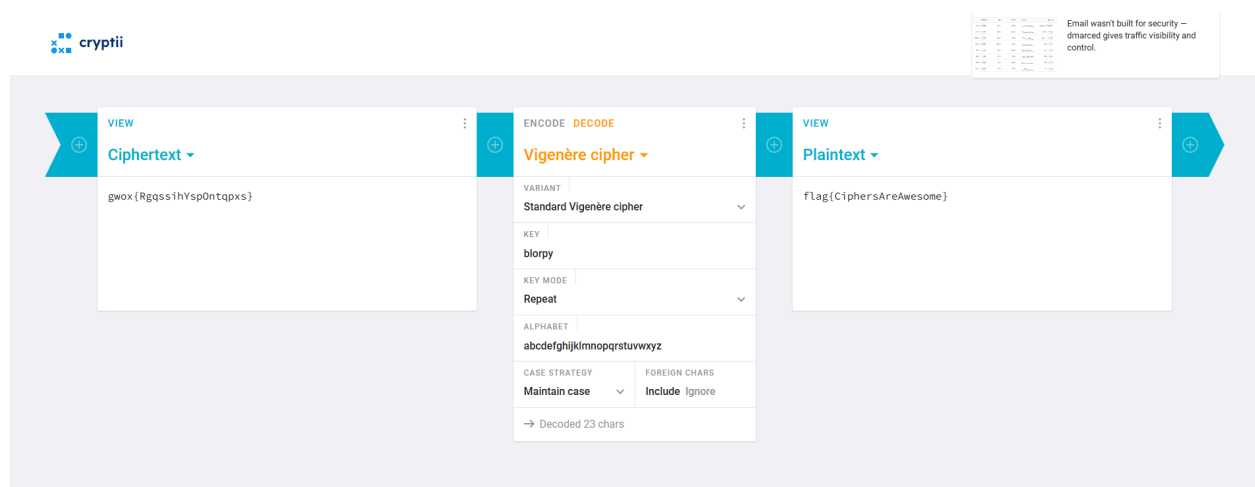
I'm not sure what this means, but it was left lying around: blorpy

gwox{RgqssihYspOntqpxs}

Initial Analysis: It tells us about the vigenere cipher and blorpy is the key for the cipher to decrypt so I will try that.

Tools Used: Cryptii

Solution: I went to the website and selected vigenere cipher and pasted the string and also pasted the key and it gave me the plain text.



Flag: flag{CiphersAreAwesome}

Takeaway: The Polyalphabetic Shift: Unlike Caesar, which uses one shift, Vigenère uses a keyword to apply different shifts to each letter. It's essentially multiple Caesar ciphers layered together.

Key Repetition: The keyword repeats over the length of the plaintext. If the key is DOG and the text is HELLO, the shifts are based on DOGDO.

Known Plaintext: If you know the flag starts with CTF{, you can subtract CTF{ from the ciphertext to instantly reveal the first four letters of the keyword.

Initial Analysis: It is morse code.

Solution: I went to the cryptii website and pasted the morse code and it decoded the code and gave me the flag.



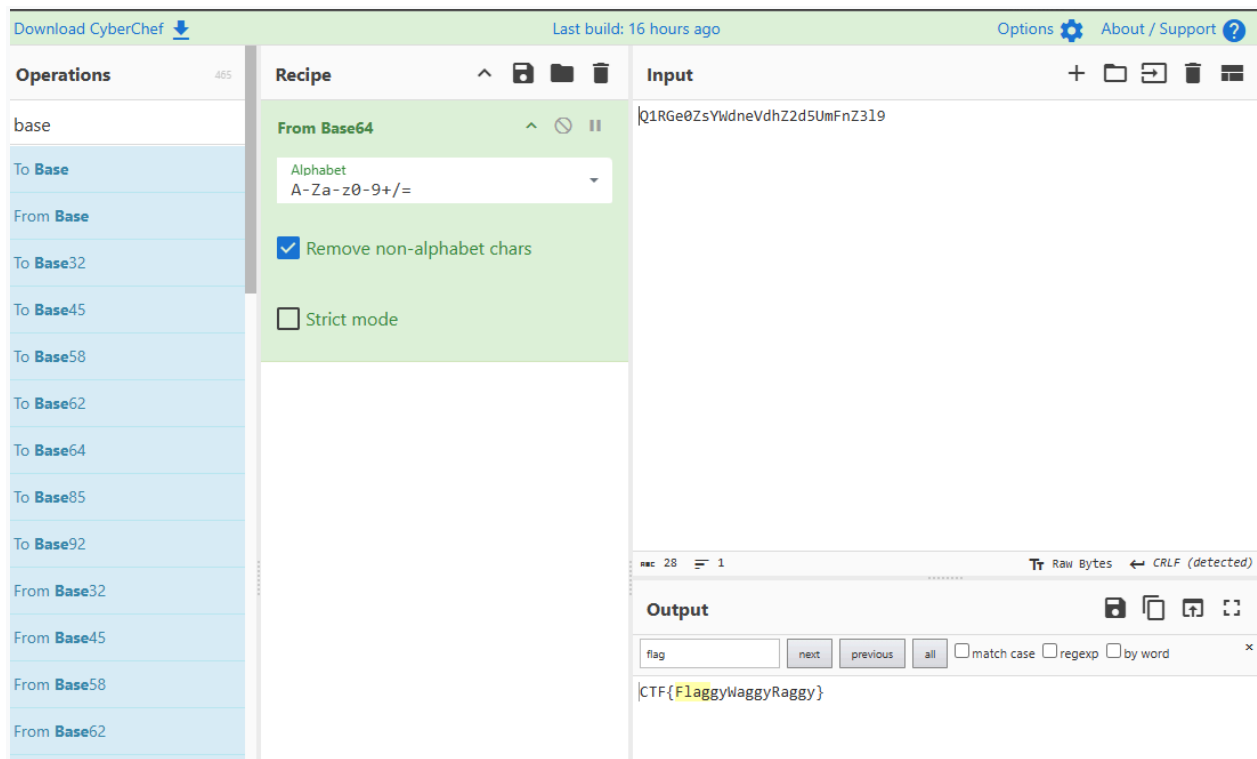
Delimiters are Key: Standard Morse uses a single space between letters and a forward slash / or triple space between words. If there are no spaces, it becomes a "Fractionated Morse" challenge harder.

Problem: There are so many different ways of encoding and decoding information nowadays... One of them will work! Q1RGe0ZsYWdneVdhZ2d5UmFnZ3l9

Initial Analysis: The problem's name Base 2 2 the 6 is Base64 so this was the hint to decode it.

Tools Used: CyberChef

Solution: I went to the cyberchef website and selected from BASE64 and pasted the cipher and it gave me the flag.



Flag:CTF{FlaggyWaggyRaggy}

Takeaway: The Character Set: Easily spotted by the use of A–Z, a–z, 0–9, and the + and / symbols. The Padding "Tell": The most iconic feature is the trailing equals signs (= or ==). These are used as padding to ensure the data aligns to a multiple of 4 characters.

Not Encryption: Base64 is encoding, not encryption. It's used to transport binary data (like images or files) as plain text. It offers zero security; anyone can decode it instantly.

Problem: A company stored a secret message on a server which got breached due to the admin using weakly hashed passwords. Can you gain access to the secret stored within the server?

```
picoCTF Webshell Help [?] [↩] [↻]

=====
Welcome to the picoCTF webshell!

❏ The webshell is intended only for solving picoCTF challenges. Any
   other usage is a violation of our terms and conditions.

📧 Sessions are monitored and logged to prevent abuse. Please do not
   enter any sensitive information into the webshell.

📁 Files stored outside of your home directory will not persist between
   webshell sessions.

🌐 Network connectivity and resources are limited. Some limits can be
   checked by typing usage.

🕒 Idle sessions will automatically log out after 15 minutes.

📖 For more information and a beginner's guide, type less ~/README.txt.
=====

KArmAkeR-picoctf@webshell:~$ nc verbal-sleep.picoctf.net 58882
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: 
```

Tools Used: Crackstation, [Hashes.com](https://hashes.com)

Solution: First i ran the pico instance then it gave me a hash i went to hashes and pasted that hash and i got a md5 detection and then went to crack station and cracked the hash the password was password123 and then another hash i followed the same approach and it was SHA-1 hash and cracked it its password was letmein and after that another hash it was SHA-256 and the password was qwerty098 after that i got the flag.

```
=====
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KArmAkeR-picoctf@webshell:~$ nc verbal-sleep.picoctf.net 58882
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the pa^C
KArmAkeR-picoctf@webshell:~$ ^C
KArmAkeR-picoctf@webshell:~$ nc verbal-sleep.picoctf.net 58882
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Incorrect. Goodbye.
^C
KArmAkeR-picoctf@webshell:~$ nc verbal-sleep.picoctf.net 58882
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Correct! You've cracked the MD5 hash with no secret found!

Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: letmein
Correct! You've cracked the SHA-1 hash with no secret found!

Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745
Enter the password for the identified hash: qwerty098
Correct! You've cracked the SHA-256 hash with a secret found.
The flag is: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_eb2f8459}
█
```

Flag: `picoCTF{UseStr0nG_h@shEs_&PaSswDs!_eb2f8459}`

Problem: This service provides you an encrypted flag. Can you decrypt it with just N & e?
Source code i got.

```
1  from sys import exit
2  from Crypto.Util.number import bytes_to_long, inverse
3  from setup import get_primes
4
5  e = 65537
6
7  def gen_key(k):
8      """
9      Generates RSA key with k bits
10     """
11     p,q = get_primes(k//2)
12     N = p*q
13     d = inverse(e, (p-1)*(q-1))
14
15     return [(N,e), d]
16
17 def encrypt(pubkey, m):
18     N,e = pubkey
19     return pow(bytes_to_long(m.encode('utf-8')), e, N)
20
21 def main(flag):
22     pubkey, _privkey = gen_key(1024)
23     encrypted = encrypt(pubkey, flag)
24     return (pubkey[0], encrypted)
25
26 if __name__ == "__main__":
27     flag = open('flag.txt', 'r').read()
28     flag = flag.strip()
29     N, cypher = main(flag)
30     print("N:", N)
31     print("e:", e)
32     print("cyphertext:", cypher)
33     exit()
34
35
```

```
=====
KArmAkeR-picocTF@webshell:~$ nc verbal-sleep.picocTF.net 63054
N: 1738769872383134547060779799613517931803065961835646026388268447358793541231174460170404775654706555541803474215852615974769077859773569183325911
7090849762
e: 65537
cyphertext: 1557492655850308660050574190569152659207439146152461548178115159602555133453561066319148943068474191598937874365751605819780317427057066
6339164910786373105
█
```

Tools used: <https://www.dcode.fr/rsa-cipher>

Solution: I went to the [dcode.fr](https://www.dcode.fr) and searched for rsa cipher and put those values in the respective places and got the Flag.



Flag: picoCTF{tw0_1\$_pr!m341c6ed35}

Problem: Can you get the real meaning from this file?

File consisted this:

YidkM0JxZGtwQIRYdHFhR3g2YUhsZmF6TnFIVGwzWVROclgya3INRFJvYTJvMmZR
PT0nCg==

Initial Analysis: It seems like a base64 the last two equal sign points kinda that.

Tools Used: CyberChef, Cryptii

Solution: Once i applied the base64 and i got this one

b'd3BqdkpBTXtqaGx6aHlfazNqeTl3YTNrX2kyMDRoa2o2fQ=='

Then again i applied the base64 and deleted the b and ' and it kinda shaped like my flag. Then i went to the cryptii website and pasted it and selected the caesar cipher and i got the flag.

Last build: 10 hours agoOptions

405Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars

☐ Strict mode

Input

YidkM0JxZGtwQlRYdHFhR3g2YUhsZmF6TnF1VGwzWVRoc1gya31NRfJvYTJvMmZRPT0nCG=

rec 72 1

Output

flag next previous all ☐ match case ☐ regexp ☐ by word

b'd3BqdkpBTXtqaGx6aH1fazNqeT13YTNrX2kyMDRo2o2fQ=

Last build: 16 hours ago

Input

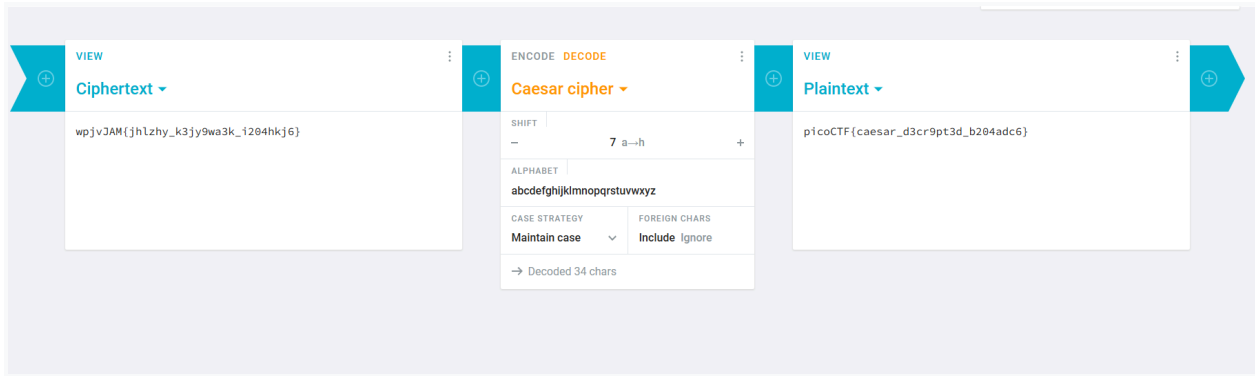
d3BqdkpBTXtqaGx6aH1fazNqeT13YTNrX2kyMDRo2o2fQ=

rec 48 1

Output

flag next previous all ☐ match case ☐ regexp ☐ by word

wpjvJAM{jh1zhy_k3jy9wa3k_i204hkj6}



Flag:`picoCTF{caesar_d3cr9pt3d_b204adc6}`

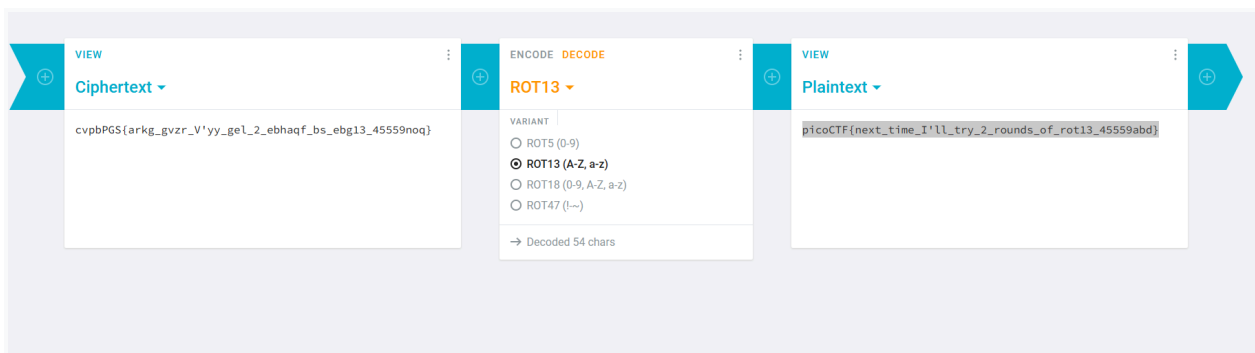
Problem: Cryptography can be easy, do you know what ROT13 is?

Values.txt file contains:`cvpbPGS{arkg_gvzr_V'yy_gel_2_ebhaqf_bs_ebg13_45559noq}`

Initial Analysis: ROT13 is our hint here. I'm gonna use it to decipher it.

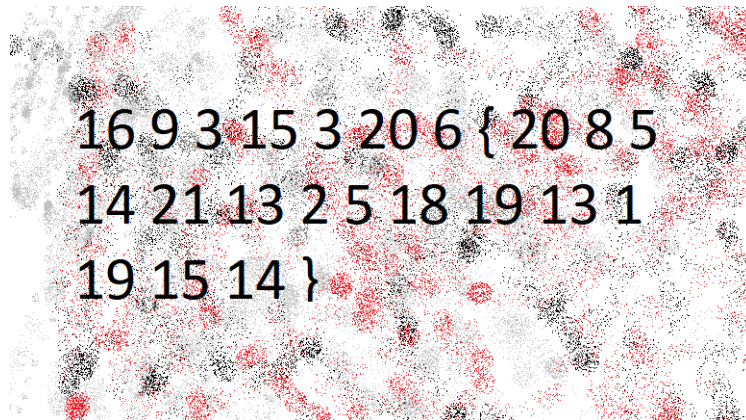
Tools used: Cryptii

Solution: I went to the cryptii website and selected decode and ROT13 and pasted the cipher text and it decoded the cipher and I got my flag.



Flag:`picoCTF{next_time_I'll_try_2_rounds_of_rot13_45559abd}`

Problem: The numbers... what do they mean?



Initial Analysis: after close inspection we can see these are alphabet numbers so it could be a letter number cipher.

Tools Used: <https://www.dcode.fr/letter-number-cipher>

Solution: I went to the [dcode.fr](https://www.dcode.fr) website and pasted the number and it revealed the flag.



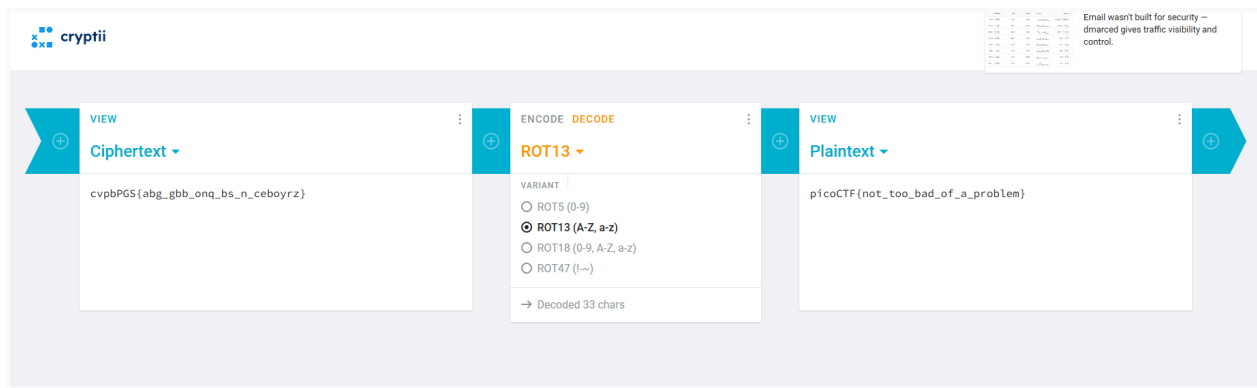
Flag: PICOCTF{THENUMBERSMASON}

Problem: Cryptography can be easy, do you know what ROT13 is?
cvpbPGS{abg_gbb_onq_bs_n_ceboyrz}

Initial Analysis: There is the hint ROT13, i will go with that.

Tools Used: Cryptii

Solution: I went to the cryptii website and selected the ROT13 and pasted the cipher text and got the flag.



Flag: picoCTF{not_too_bad_of_a_problem}

Problem: Try to decrypt the secret cheese password to prove you're not the imposter!

Tools Used: Dcode.fr

```
*****
***          Part 1          ***
***  The Mystery of the CLONED RAT  ***
*****
```

The super evil Dr. Lacktoes Inn Tolerant told me he kidnapped my best friend, S queexy, and replaced him with an evil clone! You look JUST LIKE SQUEEXY, but I' m not sure if you're him or THE CLONE. I've devised a plan to find out if YOU'R E the REAL SQUEEXY! If you're Squeexy, I'll give you the key to the cloning roo m so you can maul the imposter...

Here's my secret cheese -- if you're Squeexy, you'll be able to guess it: RSBF THPGEI

Hint: The cheeses are top secret and limited edition, so they might look differ ent from cheeses you're used to!

Commands: (g)uess my cheese or (e)ncrypt a cheese

What would you like to do?

e

What cheese would you like to encrypt? CHEDDAR

Here's your encrypted cheese: RSHMMBU

Not sure why you want it though...*squeak* - oh well!

I don't wanna talk to you too much if you're some suspicious character and not my BFF Squeexy!

You have 2 more chances to prove yourself to me!

Commands: (g)uess my cheese or (e)ncrypt a cheese

What would you like to do?

g

```
(q\_/_p)
/. .\.-.....-
=\_t\_/= / \ (____
)\ \ )__--__\ |____)
(/-(/^\nn---'
```

SQUEAK SQUEAK SQUEAK

```
(q\_/_p)
/. .\
'__ =\_t\_/=
) / \
```

```

  \  /\) (/ \
  ^- \  Y  /
    nn^nn

```

Is that you, Squeexy? Are you ready to GUESS...MY...CHEEEEEEESE?
 Remember, this is my encrypted cheese: RSBFTHPGEI
 So...what's my cheese?
 CHAUMESZPJ

```

      (q\_ /p)
      / . . \
  ,--- =\_ t\_ / =
  )      /      \
  (      ((      ))
  \      /\) (/ \
  ^- \  Y  /
    nn^nn

```

```

      . ' o 0' - .
      / 0 o - - - |
      / 0 - - - 0 |
      | o   o o |
      | o   o 0 - |
      | 0 - - - '
      - - -

```

munch...

```

      (q\_ /p)
      / . . \
  ,--- =\_ t\_ / =
  )      /      \
  (      ((      ))
  \      /\) (/ \
  ^- \  Y  /
    nn^nn

```

```

      . ' o 0' - .
      / 0 o - - - |
      ) - - - 0 |
      )   o o |
      | o   o 0 - |
      | 0 - - - '
      - - -

```

munch...

```

      (q\_ /p)
      / . . \
  ,--- =\_ t\_ / =
  )      /      \
  (      ((      ))
  \      /\) (/ \
  ^- \  Y  /
    nn^nn

```

```

      . ' o 0' - .
      / 0 o - - - |
      ) ' 0 |
      ) o |
      ) 0 - - '
      ) - - - '
      - - -

```

MUNCH.....

YUM! MMMMMMMMMMMMMMMMMM!!! Yes...yesssss! That's my cheese!
 Here's the password to the cloning room: picoCTF{ChEeSydd15526f}
 []

Solution: This is an affine cipher and it actually calculates the values and encrypts it. I used [dcode.fr](https://dcode.fr/affine-cipher) to get the A and B coefficients to get the exact values to decipher it.

Flag:picoCTF{ChEeSydd15526f}